# UMAP projections & cell-type composition

Nandhu_version_final · 233,949 cells · 49 ROIs · 11 patients (6 non-relapse, 5 relapse) · 2026-11-11

## Method

• **Input.** The 25 selected markers in final_adata.X, double z-normalised per machine. These are the same values the cell types were annotated from. **No clustering was run.** Every cell keeps its Nandhu_version_final label.

• **Batch correction for the picture only.** On the uncorrected UMAP, Machine A and Machine B cells fall in largely separate halves (fig 4, top). Machine is partly tied to outcome (relapse ROIs are 22 Machine A + 4 Machine B), so an uncorrected UMAP would partly show machine instead of biology. The 15 principal components were therefore Harmony-corrected for machine before UMAP (umap-learn, 15 neighbours, min_dist 0.3, seed 0). After correction the two machines overlap (fig 4, bottom). Both embeddings are saved in umap_coordinates.csv.

• **Composition statistics.** Each cell type's % of all cells is compared per ROI (23 vs 26) and per patient (each patient = mean of its ROIs, 6 vs 5), using Mann–Whitney with BH across cell types.

## Key observations

• **The cell types form coherent, separate regions** on the corrected UMAP (fig 1): granulocytes, erythroid, endothelium, T cells, plasma cells and mast cells each occupy their own area, and the marker plots (fig 5) line up with the labels. NPM1 blasts appear in two regions, consistent with heterogeneous blasts: the recovered tumour includes the former 'leukemic erythroid' and immature/monocytic blast groups of v2.

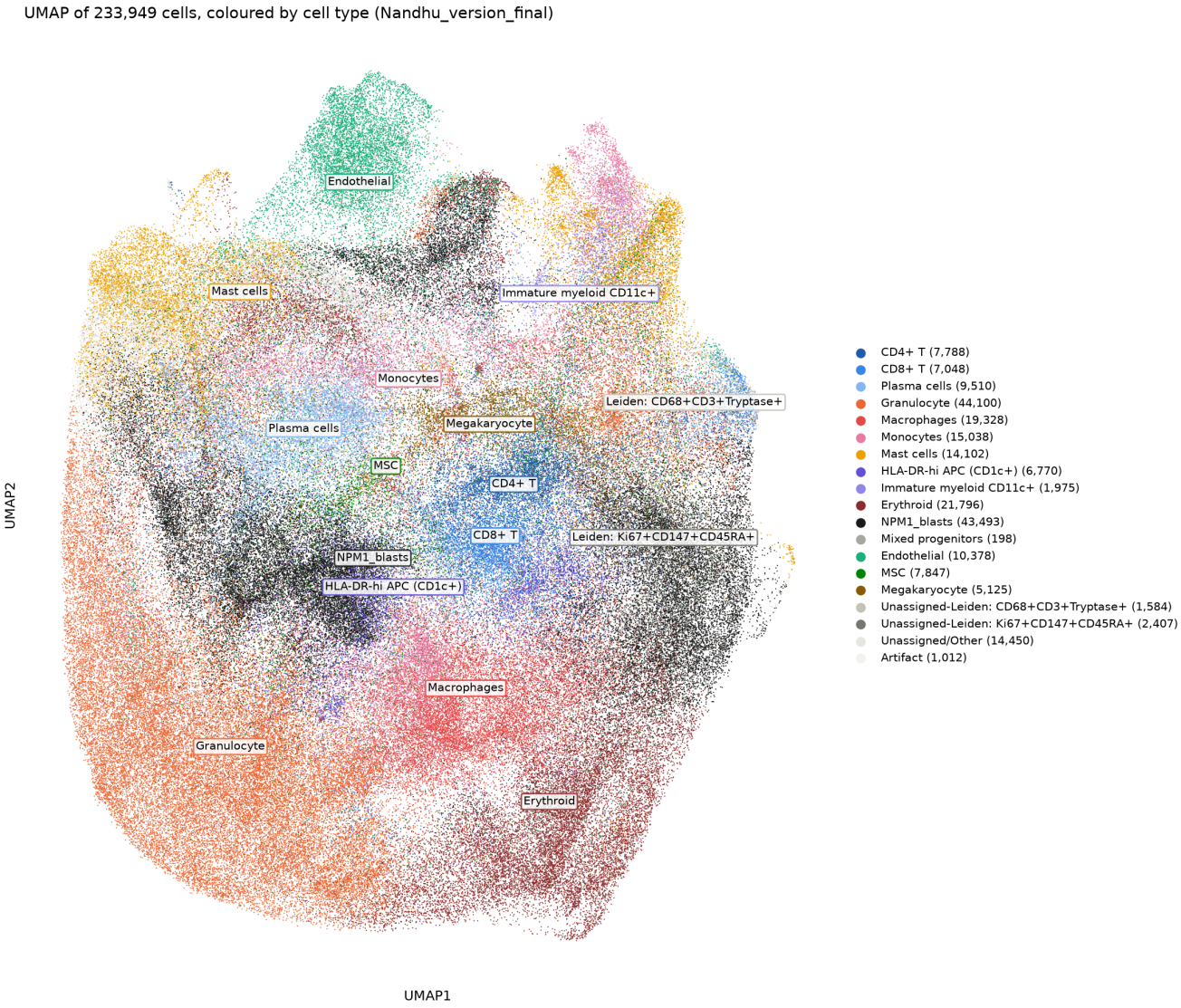
• **Relapse cells pile up in the blast regions; non-relapse cells in normal-haematopoiesis regions** (figs 2–3): granulocytic, erythroid, endothelial and mast-cell areas. The differences are in which regions are populated, not in where the cell types sit.

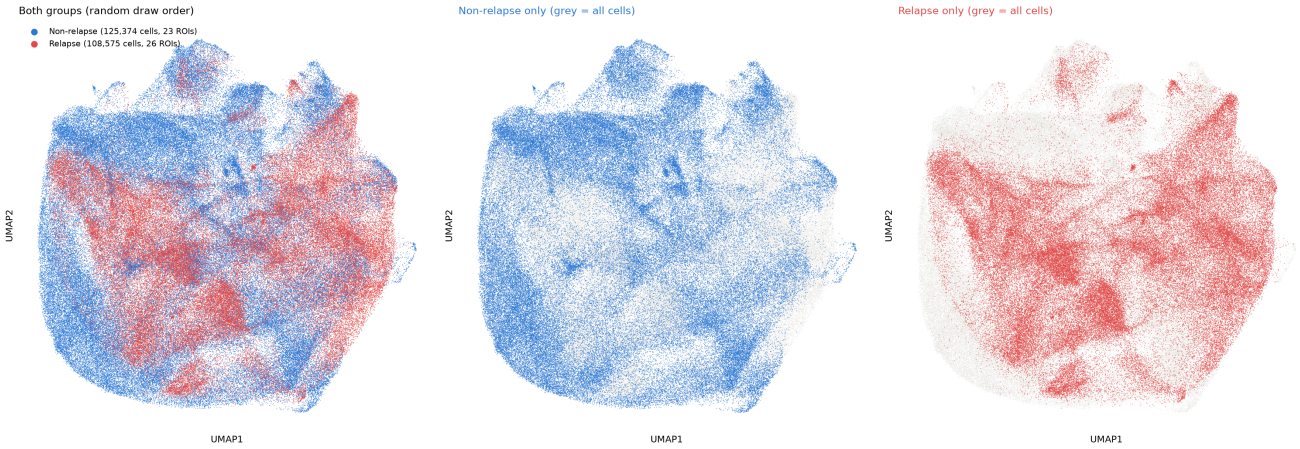
• **Two composition differences hold at both ROI and patient level:** NPM1 blasts (about 9% vs 27%; patient BH 0.041) and endothelial cells (about 9% vs 2%; patient BH 0.041). Everything else, including CD4/CD8 T cells, macrophages and monocytes, is not different per patient (figs 7, 9).

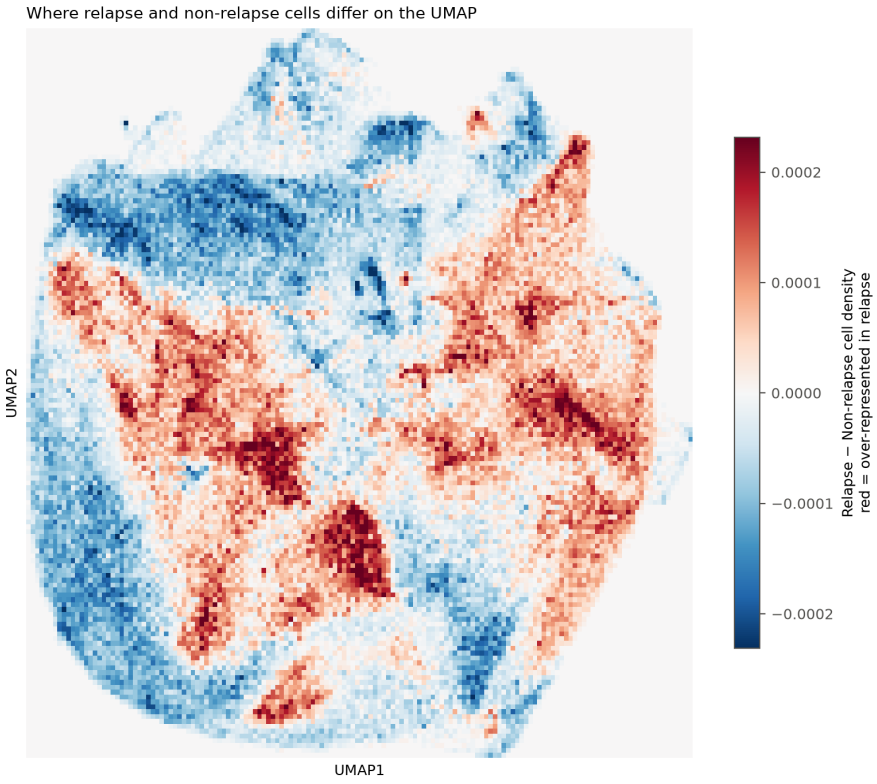
• **Large differences between patients** (fig 8). 'Immature myeloid CD11c+' is concentrated in two non-relapse patients (NR-P3 15.8%, NR-P4 9.0%). R-P4 (2 ROIs) is dominated by plasma cells. NR-P6 is CD8+ T rich. These patient-specific features explain why several ROI-level differences do not survive per patient.

## Patients

| Patient | Group | Stratification | Acquisition | ROIs | Cells |
|---|---|---|---|---|---|
| NR-P1 | Non-relapse | CR MRD+ no relapse | NPM1_MRD_06_2024 | 4 | 29,443 |
| NR-P2 | Non-relapse | MRD+ no relapse | NPM1_MRD_04_2025 | 7 | 47,370 |
| NR-P3 | Non-relapse | CR MRD+ no relapse | NPM1_MRD_06_2024 | 3 | 7,508 |
| NR-P4 | Non-relapse | CR MRD+ no relapse post HCT | NPM1_MRD_02_2025 | 2 | 6,388 |
| NR-P5 | Non-relapse | PR MRD+ no relapse post allo; in sample CR | NPM1_MRD_06_2024 | 4 | 30,277 |
| NR-P6 | Non-relapse | CR MRD+ no relapse | NPM1_MRD_02_2025 | 3 | 4,388 |
| R-P1 | Relapse | CR MRD+ relapse | NPM1_MRD_06_2024 | 4 | 10,838 |
| R-P2 | Relapse | CR MRD+ relapse | NPM1_MRD_04_2025 | 9 | 41,341 |
| R-P3 | Relapse | CR MRD+ relapse | NPM1_MRD_02_2025 | 7 | 16,755 |
| R-P4 | Relapse | CR MRD+ relapse | NPM1_MRD_04_2025 | 2 | 7,838 |
| R-P5 | Relapse | CR MRD+ relapse | NPM1_MRD_02_2025 | 4 | 31,803 |

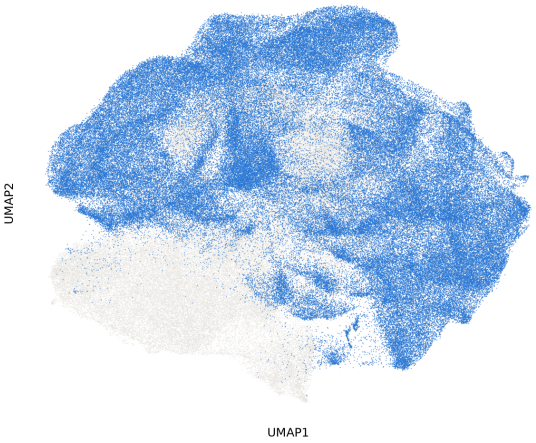

UMAP of 233,949 cells, coloured by cell type (Nandhu_version_final)
Endothelial
Mast cells
Immature myeloid CD11c+
Monocytes
Leiden: CD68+CD3+Tryptase+
Plasma cells
Megakaryocyte
MSC
CD4+ T
CD8+ T
Leiden: Ki67+CD147+CD45RA+
NPM1_blasts
HLA-DR-hi APC (CD1c+)
Macrophages
Granulocyte
Erythroid
UMAP1
UMAP2
CD4+ T (7,788)
CD8+ T (7,048)
Plasma cells (9,510)
Granulocyte (44,100)
Macrophages (19,328)
Monocytes (15,038)
Mast cells (14,102)
HLA-DR-hi APC (CD1c+) (6,770)
Immature myeloid CD11c+ (1,975)
Erythroid (21,796)
NPM1_blasts (43,493)
Mixed progenitors (198)
Endothelial (10,378)
MSC (7,847)
Megakaryocyte (5,125)
Unassigned-Leiden: CD68+CD3+Tryptase+ (1,584)
Unassigned-Leiden: Ki67+CD147+CD45RA+ (2,407)
Unassigned/Other (14,450)
Artifact (1,012)


Both groups (random draw order)
Non-relapse (125,374 cells, 23 ROIs)
Relapse (108,575 cells, 26 ROIs)
Non-relapse only (grey = all cells)
Relapse only (grey = all cells)
UMAP2
UMAP1

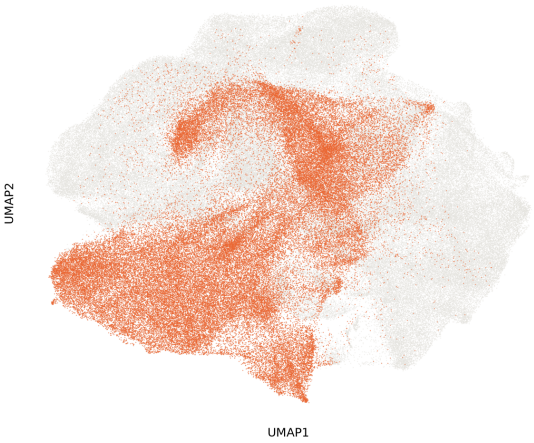

Where relapse and non-relapse cells differ on the UMAP
UMAP2
UMAP1
0.0002
0.0001
0.0000
−0.0001
−0.0002
Relapse − Non-relapse cell density
red = over-represented in relapse

Batch check: cells from each machine before and after Harmony (grey = all cells)

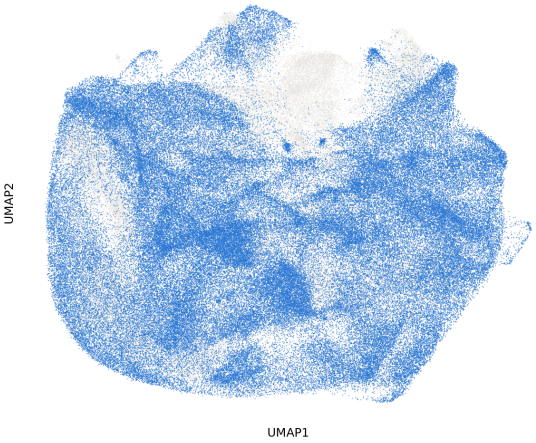
uncorrected PCs
machine_A (155,883 cells)



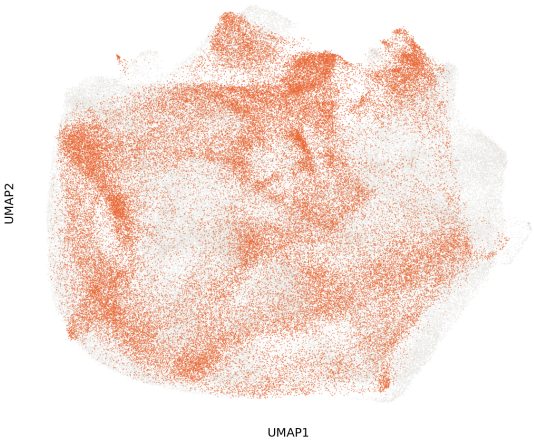

uncorrected PCs
machine_B (78,066 cells)



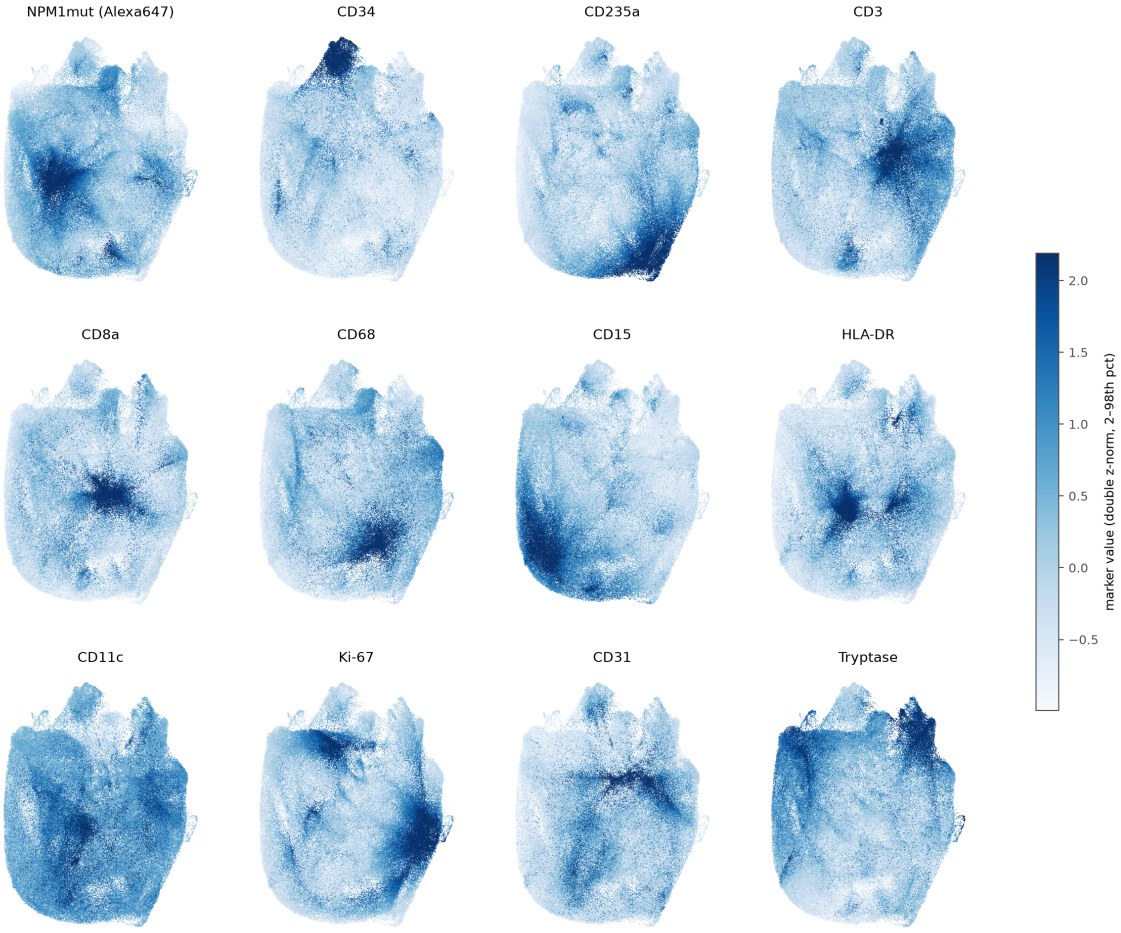

Harmony-corrected PCs (used in all other figures)
machine_A (155,883 cells)



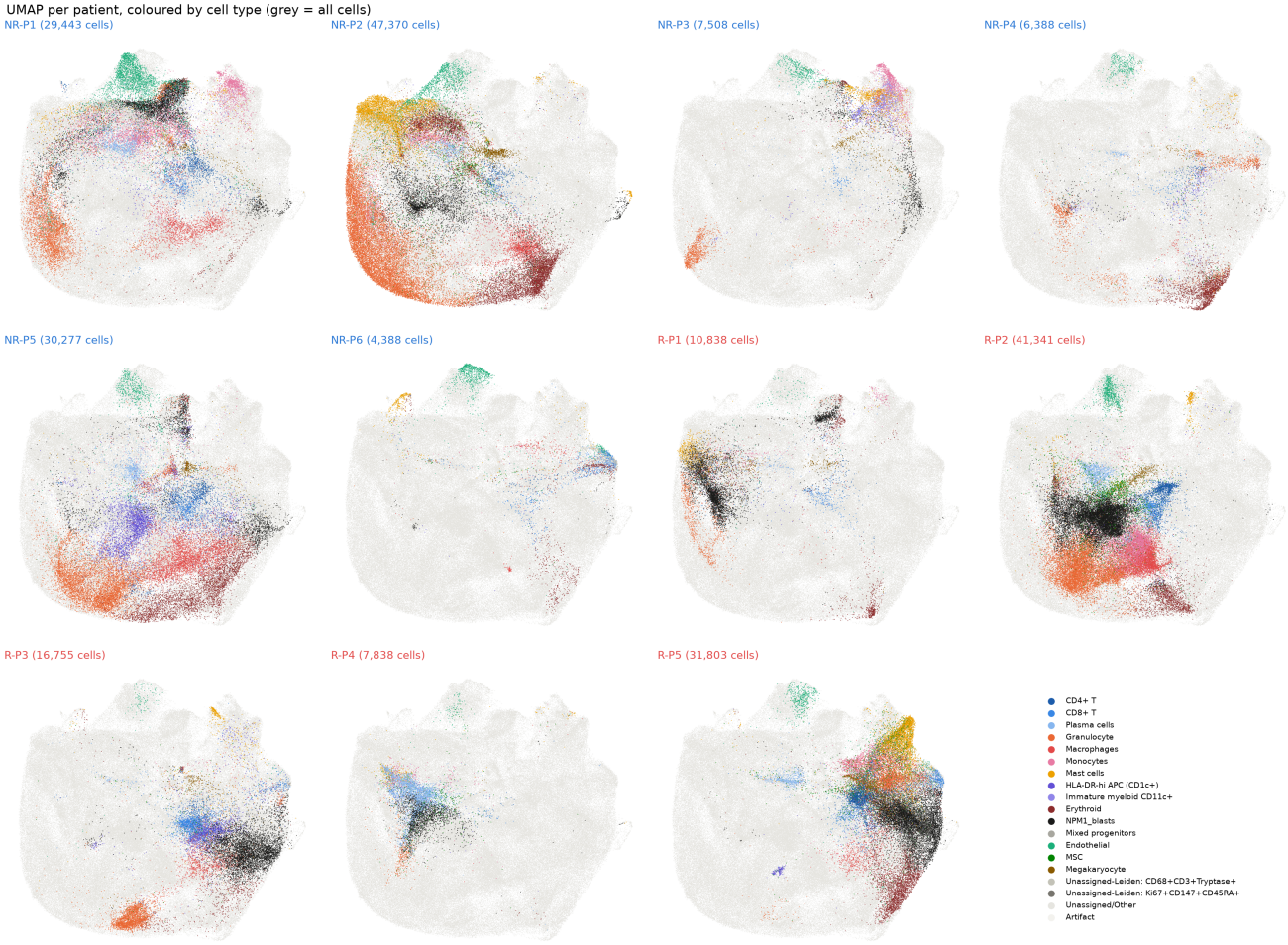

Harmony-corrected PCs (used in all other figures)
machine_B (78,066 cells)



Marker expression on the UMAP (high values drawn on top)

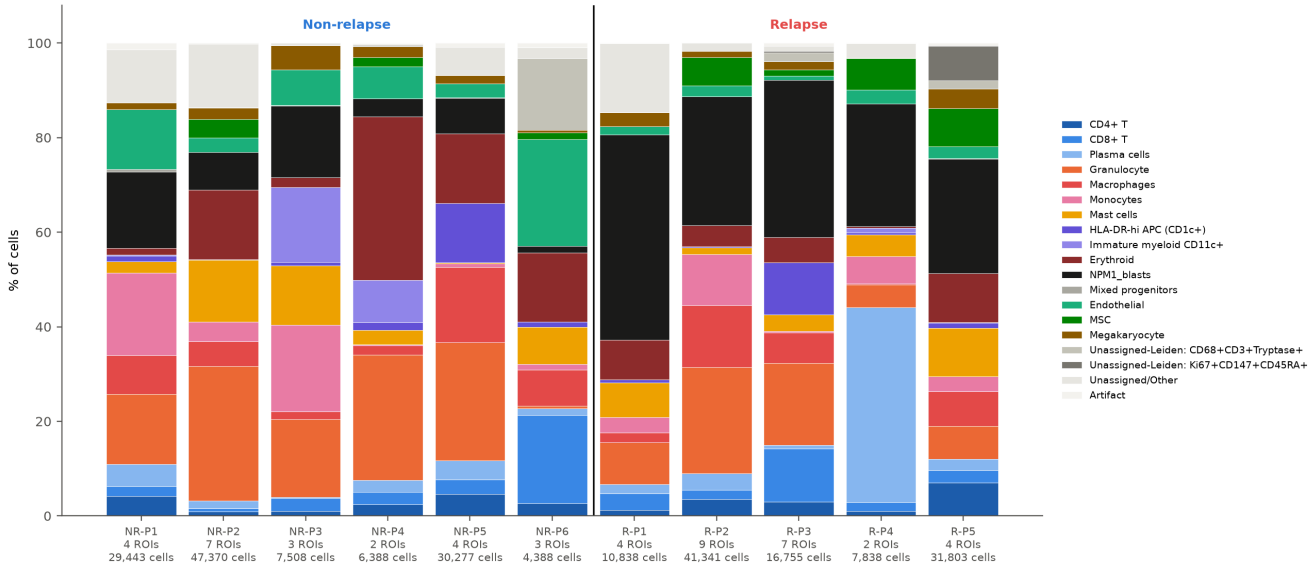


UMAP per patient, coloured by cell type (grey = all cells)

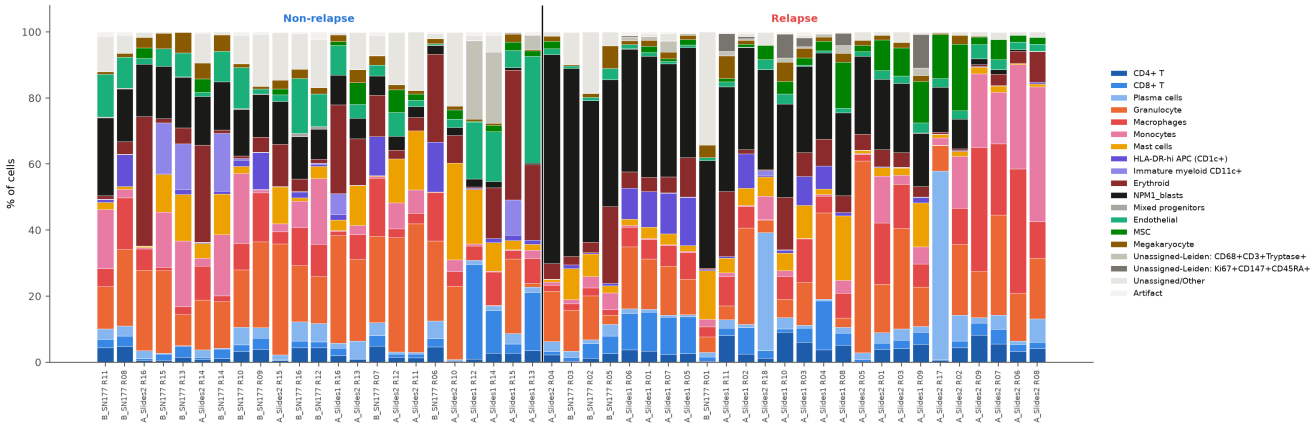


# Cell-type composition

Cell-type composition per patient (all cells of the patient's ROIs pooled)

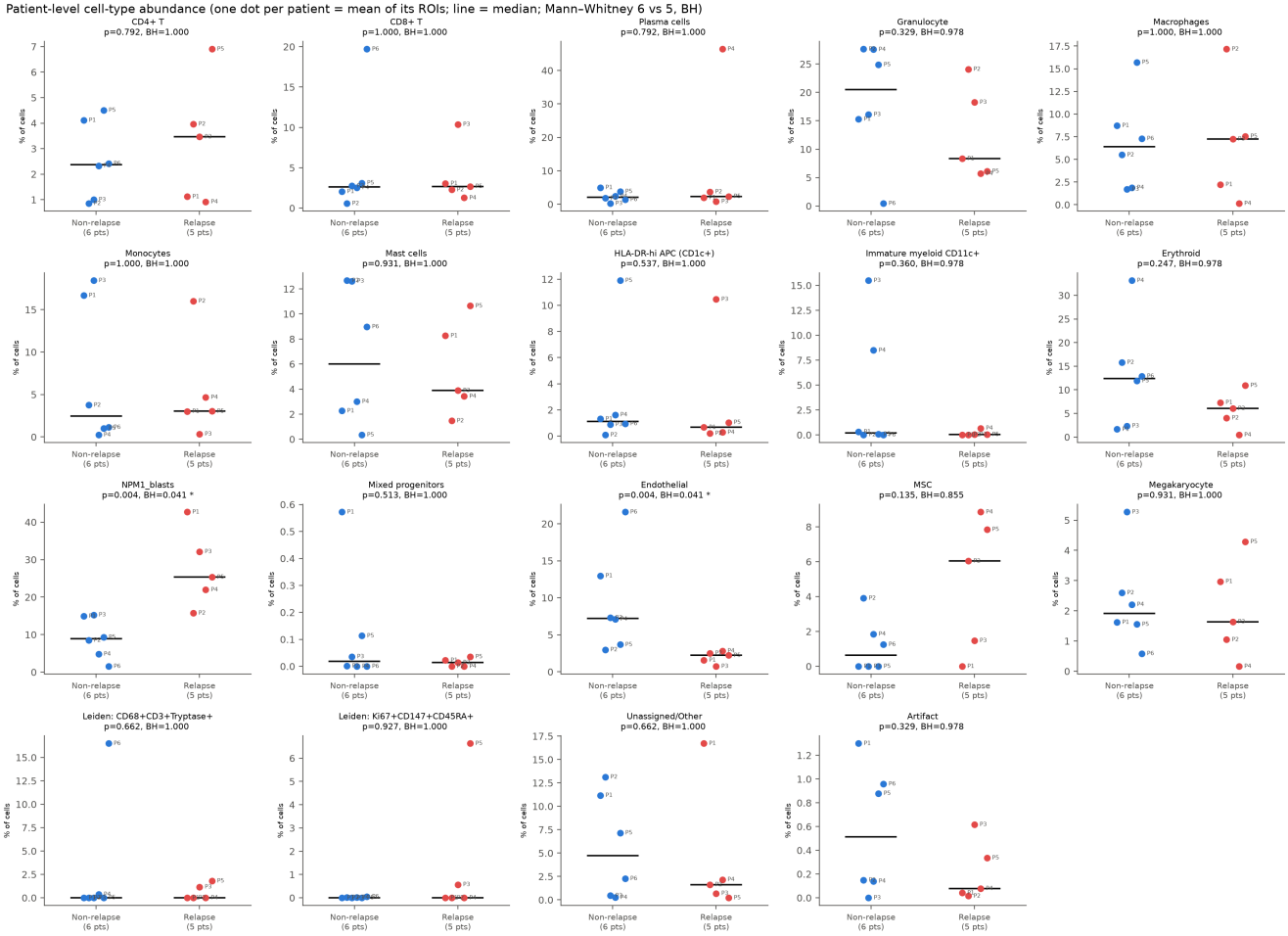


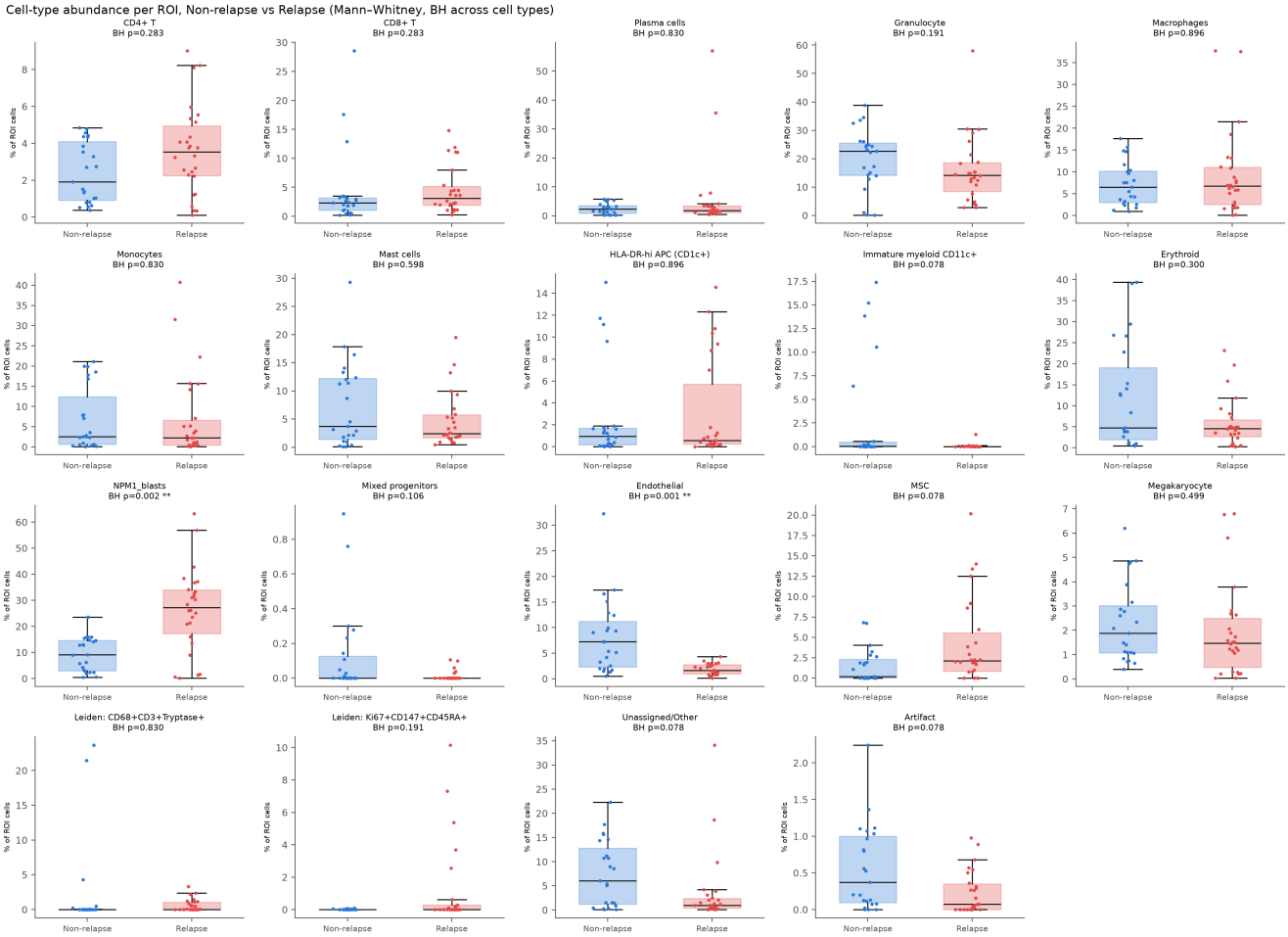

Cell-type composition per ROI (sorted by % NPM1 blasts within each group)



| Cell type | NR % (ROI) | R % (ROI) | ROI BH p | NR % (pt) | R % (pt) | patient p | patient BH |
|---|---|---|---|---|---|---|---|
| CD4+ T | 2.4 | 3.6 | 0.283 | 2.5 | 3.3 | 0.792 | 1.000 |
| CD8+ T | 4.2 | 4.6 | 0.283 | 5.1 | 3.9 | 1.000 | 1.000 |
| Plasma cells | 2.5 | 5.8 | 0.830 | 2.5 | 11.1 | 0.792 | 1.000 |
| Granulocyte | 20.0 | 15.9 | 0.191 | 18.7 | 12.5 | 0.329 | 0.978 |
| Macrophages | 7.3 | 9.4 | 0.896 | 6.8 | 6.9 | 1.000 | 1.000 |
| Monocytes | 6.8 | 6.9 | 0.830 | 6.9 | 5.4 | 1.000 | 1.000 |
| Mast cells | 7.4 | 4.7 | 0.598 | 6.6 | 5.5 | 0.931 | 1.000 |
| HLA-DR-hi APC (CD1c+) | 2.7 | 3.2 | 0.896 | 2.8 | 2.5 | 0.537 | 1.000 |
| Immature myeloid CD11c+ | 2.8 | 0.1 | 0.078 | 4.1 | 0.2 | 0.360 | 0.978 |
| Erythroid | 12.0 | 5.9 | 0.300 | 13.0 | 5.8 | 0.247 | 0.978 |
| NPM1_blasts | 9.4 | 26.3 | 0.002 | 9.1 | 27.6 | 0.004 | 0.041 |
| Mixed progenitors | 0.1 | 0.0 | 0.106 | 0.1 | 0.0 | 0.513 | 1.000 |
| Endothelial | 8.2 | 1.9 | 0.001 | 9.3 | 2.0 | 0.004 | 0.041 |
| MSC | 1.5 | 4.4 | 0.078 | 1.2 | 4.8 | 0.135 | 0.855 |
| Megakaryocyte | 2.3 | 1.9 | 0.499 | 2.3 | 2.0 | 0.931 | 1.000 |
| Unassigned-Leiden: CD68+CD3+Tryptase+ | 2.2 | 0.6 | 0.830 | 2.8 | 0.6 | 0.662 | 1.000 |
| Unassigned-Leiden: Ki67+CD147+CD45RA+ | 0.0 | 1.2 | 0.191 | 0.0 | 1.4 | 0.927 | 1.000 |
| Unassigned/Other | 7.6 | 3.5 | 0.078 | 5.7 | 4.3 | 0.662 | 1.000 |
| Artifact | 0.6 | 0.2 | 0.078 | 0.6 | 0.2 | 0.329 | 0.978 |

Patient-level cell-type abundance (one dot per patient = mean of its ROIs; line = median; Mann–Whitney 6 vs 5, BH)






## Interpretation

At the whole-tissue level, relapse and non-relapse MRD marrow differ mainly in two things. Relapse marrow carries about three times more NPM1-mutant blasts, and non-relapse marrow has about four times more endothelium. The phenotypes of the other cell types are largely shared. Relapse is therefore marked by the expansion of the blast compartment and the loss of the vascular niche, rather than by new cell states. This agrees with the CN and proximity analyses. The UMAP also shows large differences between individual patients, so with 6 vs 5 patients only effects as large as these two reach significance. Smaller shifts need a larger cohort.

## Files

**umap_graphs.py / UMAP_Graphs.ipynb** is the full analysis (identical code). **umap_coordinates.csv** has per-cell coordinates (Harmony-corrected and uncorrected). The composition CSVs cover per ROI, per patient, and group statistics. The patient mapping is **../roi_patient_mapping.csv**.